

Data Analytics Capstone Topic Approval Form

Student Name: xxxxxx

Student ID: xxxxxxxx

Capstone Project Name: Predicting if the vehicle's features influence the sales volume for BMW vehicles between 2010 and 2024

Project Topic:

The project will utilize random forest regressor machine learning model to predict if the sales volume for the BMW vehicles is influenced by vehicle features. The data will be obtained from Kaggle website

(<https://www.kaggle.com/datasets/ahmadrazakashif/bmw-worldwide-sales-records-20102024> .)

☒ **This project does not involve human subjects research and is exempt from WGU IRB review.**

Research Question:

Can we utilize the random forest regressor model to predict the factors influencing BMW Sales volume between 2010 and 2024 across various regions?

Hypothesis:

Null hypothesis- The sales volume for the various BMW vehicles is not significantly influenced by the various factors such as the model, color, fuel_type, transmission, Engine_Size_L, Mileage_KM, Price_USD in the various regions from the year 2010 to the year 2024. **Alternate Hypothesis-** The sales volume for the various BMW vehicles is significantly influenced by the various factors such as the model, color, fuel_type, transmission, Engine_Size_L, Mileage_KM, Price_USD in the various regions from the year 2010 to the year 2024

Context:

The main goal of carrying out this data analysis is to determine if the features of vehicles influence the sales volume of vehicles. I will be build a random forest regressor machine learning model and use to predict if the vehicle features

influence the sales volume of the vehicle. The benefit of this data analysis is that the motor vehicle sales industry can use it to determine if they can increase vehicle sales through vehicle features. .

Data:

The BMW worldwide sales records 2010 to 2024 data that will be utilized for this data analysis was obtain from the Kaggle.com website on this link <https://www.kaggle.com/datasets/ahmadrazakashif/bmw-worldwide-sales-records-20102024>

The BMW worldwide sales records 2010 to 2024 dataset has 50000 sales records from the year 2010 the data has the features below that will be utilized for analysis.

1. Sales Volume

The variable Sales Volume is a numeric variable and will be used as a response (dependent) variable. We shall be predicting the Sales Volume of the BMW vehicles

2. Model

The variable "Model" is categorical and will be used as an independent variable.

3. Year

The variable "Year" is numerical, and it will be used as a control variable.

4. Region

The variable "Region" is categorical, and it will be used as a control variable.

5. Color

The variable "Color" is categorical and will be used as an independent variable.

6. Fuel Type

The variable "fuel Type" is categorical and will be used as an independent variable.

7. Transmission

The variable "Transmission" is categorical and will be used as an independent variable.

8. Engine_Size_L

The variable "Engine_Size_L" is numerical and will be used as an independent variable.

9. Mileage_KM

The variable "Mileage_KM" is numerical and will be used as an independent variable.

10. Price_USD

The variable "Price_USD" is numerical and will be used as an independent variable.

Explain who owns the data and why you are allowed to use this data for your capstone project.

There are no restrictions on using the dataset BMW sales data (2010-2024). Csv. The data Owner listed on the dataset that I will be using is Ahmad Raza Kashif (Owner) and the dataset has ODC Public Domain and dedication and License (PDDL) a license that allows for freely share, modify and use the work.

Data Gathering:

The data already exists on the Kaggle.com website and is available for download into a CSV file. I will download the dataset into a CSV file and then load it into a Python dataframe, from which I will explore the data and perform initial preprocessing steps to make the data usable for my model.

Data Analytics Tools and Techniques:

For this data analysis, I will be utilizing the Python tool and the Random Forest regressor technique.

The following Python libraries will be utilized to perform the various functionalities:

1. Pandas

Pandas will import the data from the CSV file, conduct exploratory data analysis, manipulate data, and convert categorical data to numerical data.

2. Numpy

Numpy will be used for data manipulation.

3. Matplotlib.pyplot

Matplotlib will be used for data visualization.

4. Seaborn

Seaborn will be used for data visualization in the exploratory data analysis, such as creating heatmaps.

5. `from sklearn.model_selection import train_test_split`

The function `train_test_split` from `sklearn.model_selection` will split the dataset into training and testing subsets.

6. `from sklearn import preprocessing`

The library `preprocessing` from `sklearn` will be used for data preparation tasks, such as scaling, encoding categorical variables, and normalizing data.

7. `from sklearn.feature_selection import SelectKBest`

`SelectKBest` from `sklearn.feature_selection` will select the top k features from the dataset based on statistical tests, reducing dimensionality and improving the model.

8. `from sklearn.feature_selection import f_regression`

The function `f_regression` from `sklearn.feature_selection` will be used to perform the ANOVA F-test to select statistically significant features for regression tasks.

9. `from sklearn.ensemble import RandomForestRegressor`

The library `RandomForestRegressor` from `sklearn.ensemble` will be used to implement the Random Forest algorithm for regression tasks

10. `from sklearn.model_selection import RandomizedSearchCV`

The function `RandomizedSearchCV` from `sklearn.model_selection` will perform hyperparameter tuning by testing different parameter combinations and selecting the best ones for the Random Forest model.

11. `from sklearn.metrics import mean_squared_error`

The function `mean_squared_error` from `sklearn.metrics` will be used to evaluate the performance of the regression model by measuring the average squared difference between predicted and actual values.

12. `from sklearn.metrics import r2_score`

The function `r2_score` from `sklearn.metrics` will be used to evaluate the proportion of variance in the target variable explained by the Random Forest Regressor.

13. `from sklearn.metrics import mean_absolute_error`

The `mean_absolute_error` function from `sklearn.Metrics` will be used to calculate the Mean Absolute Error (MAE) between predicted and actual values in a regression model.

14. warnings

Additional packages will be used to suppress future warnings and package changes expected from the library, ensuring a clean output during the analysis.

Justification of Tools/Techniques:

I chose the Python tool for the analysis because it is an open-source general-purpose programming language with many library packages that can be used and shared with others. Also, its flexibility and extensive library collections are valuable for performing complex calculations and analysis.

I chose the Random Forest regressor machine learning technique because I am trying to predict vehicle features that influence sales volume in various regions from 2010 to 2024. The target variable is continuous data, and the independent variables are mixed, whereby we have categorical and continuous data, and hence the random forest regressor algorithm is most suitable for this analysis since it is able to deal with mixed data types, and it will be able to capture the non-linear and complex relationships of the various variables. In addition, the algorithm is not sensitive to outliers, making it a stable and reliable tool for analyzing automotive sales data. Moreover, by utilizing the random forest regressor model, I will be able to answer the research question, as the model will provide a feature importance ranking, showing which features influence the sales volume.

Project Outcomes:

By utilizing the random forest regressor model on the BMW vehicles, I will create a model that predicts the features influencing the sales volume of BMW vehicles and provides a feature importance ranking. The model's accuracy will be tested to determine whether to reject or accept the null hypothesis.

Projected Project End Date: 11/26/2025

Sources:

D214: DataAnalyticsGraduateCapstoneProject_Task1_MSDA Capstone Topic Approval Form

BeatriceM_WGU_MSDA

The data was obtained from the Kaggle.com website on the link below

<https://www.kaggle.com/datasets/ahmadrazakashif/bmw-worldwide-sales-records-20102024>

Course Instructor Signature/Date:

- ☐ The research is exempt from an IRB Review.
- ☐ An IRB approval is in place (provide proof in appendix B).

Course Instructor's Approval Status: Approved

Date: Click here to enter a date.

Reviewed by:

Comments: Click here to enter text.